DATA SCIENCE & ARTIFICIAL INTELLIGENCE

Probability and Statistics

Statistics - 1

Lecture No. 01





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Recap of previous lecture



Topic

PROBABILITY

Topics to be Covered



Topic

Questions Based on Tree Diagram

Q. A Coin is tossed until Head appears then find the prob that odd Number of tosses will be Required to end such type of Random Experiment.

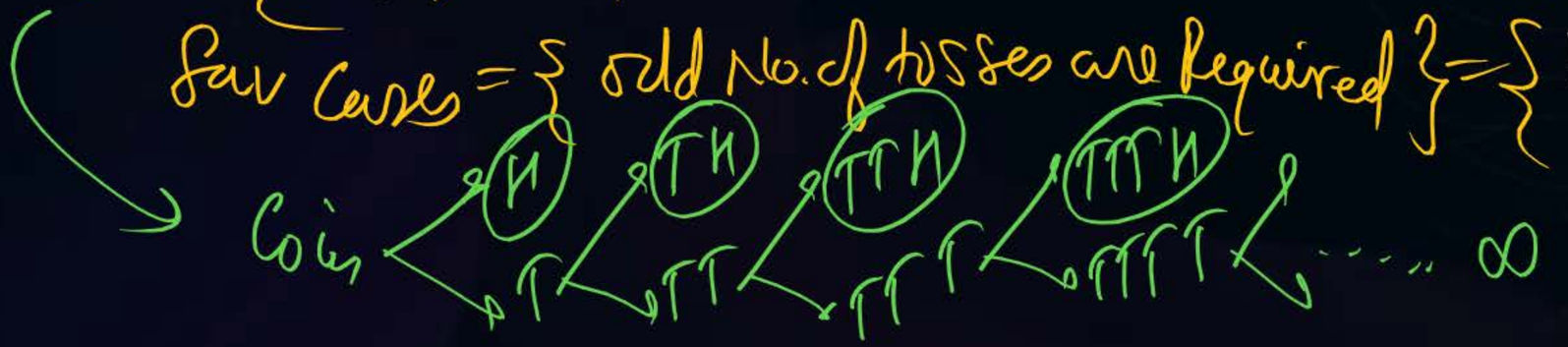
OR

Q. A Coin is tossed until Head appears then find the prob that Required No. of tosses will be odd?

Sol: $P(H)=P(T)=\frac{1}{2}$, All tosses are Independent ☺

$S=\{H, TH, TTH, TTTH, TTTTH, \ldots\}$

Fav Cases = { odd No. of tosses are Required } = $\{H, TTH, TTTTH, \ldots\}$



$$P(\text{odd tosses}) = P[H \text{ or } TTH \text{ or } TTTTH \text{ or } TTTTTTH \text{ or } \dots\dots]$$

$$= \frac{1}{2} + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^5 + \left(\frac{1}{2}\right)^7 + \dots\dots$$

$$= \frac{1}{2}\left[1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^6 + \dots\right]$$

$$= \frac{1}{2}\left[\frac{1}{1-\left(\frac{1}{2}\right)^2}\right] = \frac{1}{2}\left[\frac{4}{3}\right] = \frac{2}{3}$$ Ans

& $P(\text{even tosses}) = 1 - \frac{2}{3} = \frac{1}{3}$

$S_\infty = a + ar + ar^2 + \dots\dots$

$= \frac{a}{1-r}$

a = 1st term

r = Common Ratio

Q: Two persons A & B play a game of Die alternately in which anyone can win if 6 appears 1st time then find their respective chances of winning? if A begins.

Sol: $W = \{6 \text{ in occurs}\} \Rightarrow P(W) = \frac{1}{6}$

$L = \{6 \text{ in Not occurs}\} \Rightarrow P(L) = \frac{5}{6}$

$S = \{W, LW, LLW, LLLW, \ldots\}$

Fav Cases for $A = \{W, LLW, LLLLW, \ldots\}$



$$P(A \text{ wins}) = P[W \text{ or } LLW \text{ or } LLLLW \text{ or } \dots]$$

$$= P(W) + P(LLW) + P(LLLLW) + \dots$$

$$= \frac{1}{6} + \frac{5}{6}\cdot\frac{5}{6}\cdot\frac{1}{6} + \left(\frac{5}{6}\right)^4\cdot\frac{1}{6} + \dots$$

$$= \frac{1}{6}\left[1 + \left(\frac{5}{6}\right)^2 + \left(\frac{5}{6}\right)^4 + \dots\right]$$

$$= \frac{1}{6}\left[\frac{1}{1-\left(\frac{5}{6}\right)^2}\right] = \frac{6}{11}$$

& $P(B \text{ Win}) = 1 - \frac{6}{11} = \frac{5}{11}$

Q- Three persons A, B, C are playing a game of Die in succession till one gets six and wins the game, then find their respective chances of winning if A starts?

NWO

Ans: $P(A \text{ wins}) = \frac{36}{91}$, $P(B \text{ wins}) = \frac{30}{91}$, $P(C \text{ wins}) = 1 - \left(\frac{36}{91} + \frac{30}{91}\right) = \frac{25}{91}$

$P(W) = \frac{1}{6}$, $P(L) = \frac{5}{6}$

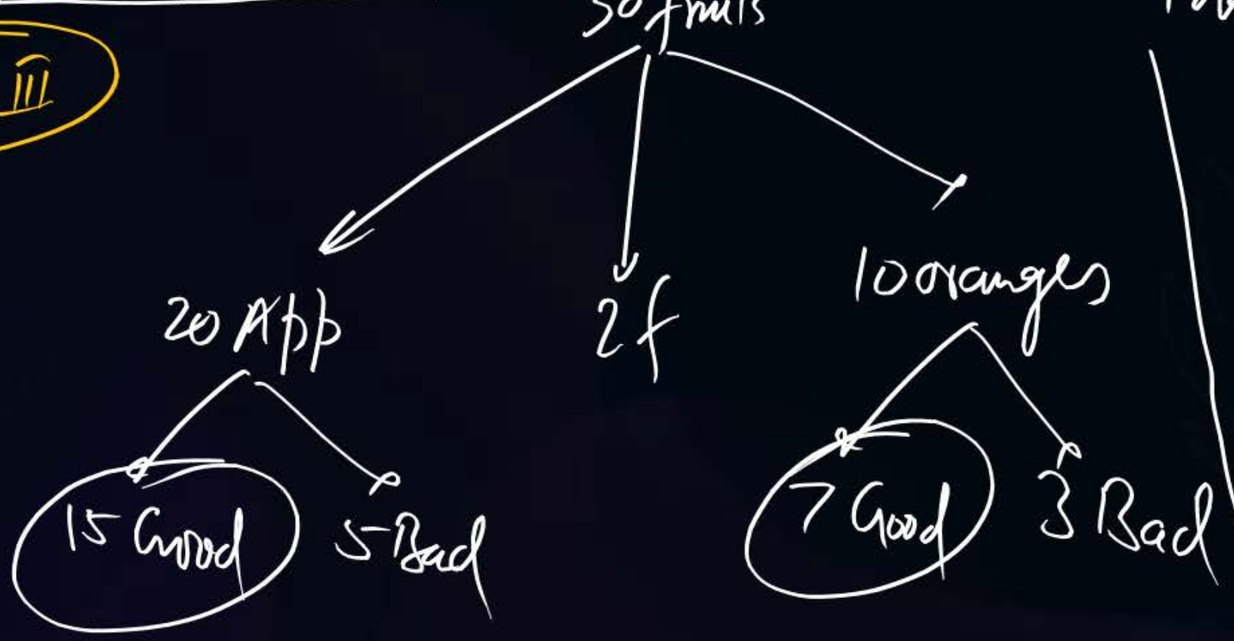


Now Do yourself.

Q: A Basket Contains 20 Apples and 10 oranges in which 5 App and 3 oranges are Rotten (=BAD). If two fruits are choosen at Random Then find the Prob that either both are Apples or Both are Good?

Sol: Information Tree:

App II & III



Total ways of Selecting 2f = $^{30}C_2 = 435$

A = {Both the selected f are Apples}

G = {" " " " are Good f}

$A \cap G$ = {Both the selected fruits are Good Apples}

$$P(A) = P(\text{Both of are Apples}) = \frac{fav}{T} = \frac{{}^{20}C_2}{{}^{30}C_2} = \frac{190}{435}$$

$$P(G) = P(\text{Both are Good f}) = \frac{fav}{T} = \frac{{}^{22}C_2}{{}^{30}C_2} = \frac{231}{435}$$

$$P(A \cap G) = P(\text{Both f are Good Apples}) = \frac{fav}{T} = \frac{{}^{15}C_2}{{}^{30}C_2} = \frac{105}{435}$$

Hence $\boxed{P(A \cup G) = P(A) + P(G) - P(A \cap G)}$

$$= \frac{190}{435} + \frac{231}{435} - \frac{105}{435} = \frac{?}{435} = 0.725$$

Qe 2010 Aishwarya studies either CS or Maths on each day.

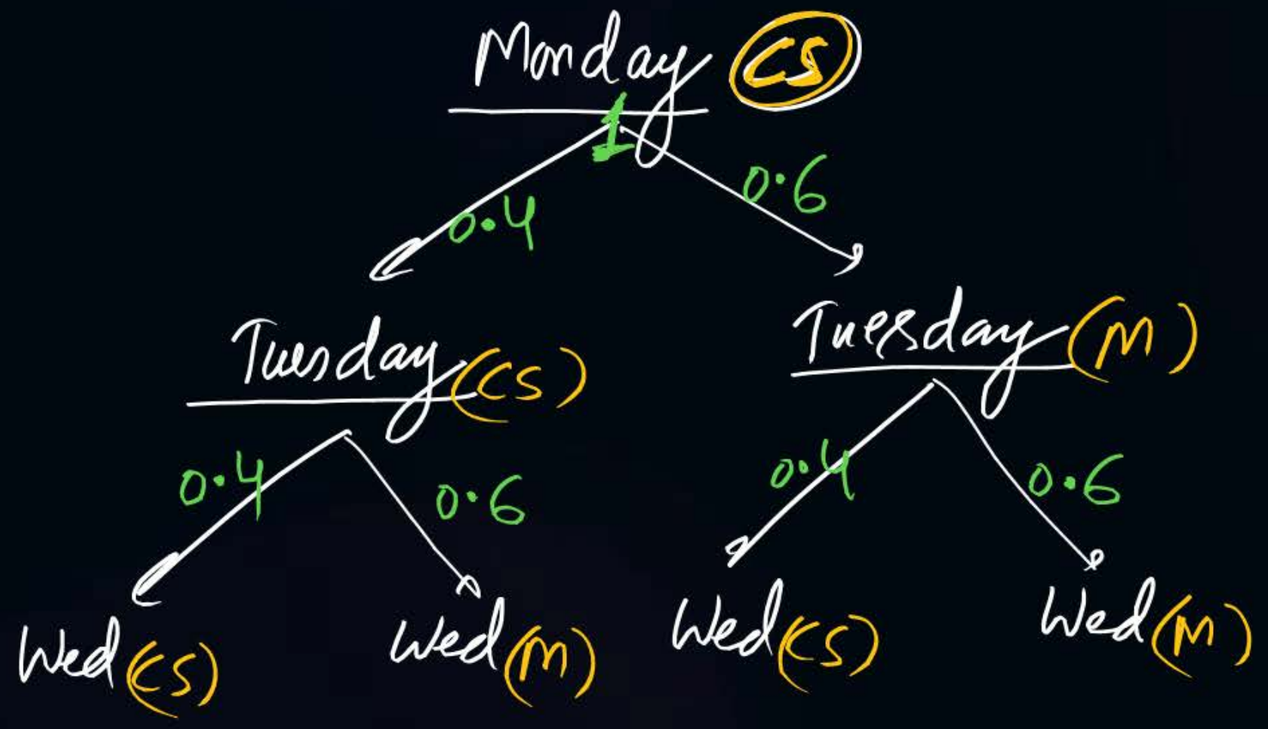
If she studies CS on a day then the Prob of studying M on next day is 0.6

& if " " M " " " " " " " " CS " " " is 0.4

Given that Aishwarya studies CS on Monday then Find the Prob that she will also study CS on Wednesday.

Sol: CS $\longrightarrow$ M ; Prob = 0.6

M $\longrightarrow$ CS ; Prob = 0.4



various possible path :

$$S=\{(CS\ CS\ CS), (CS\ CS\ M), (CS\ M\ CS), (CS\ M\ M)\}$$

E_1 E_2 E_3 E_4

Here E_1, E_2, E_3, E_4 are ME.

Fav Path $= \{(CS \text{ on Wed})\}$

$= \{(CS\ CS\ CS), (CS\ M\ CS)\}$

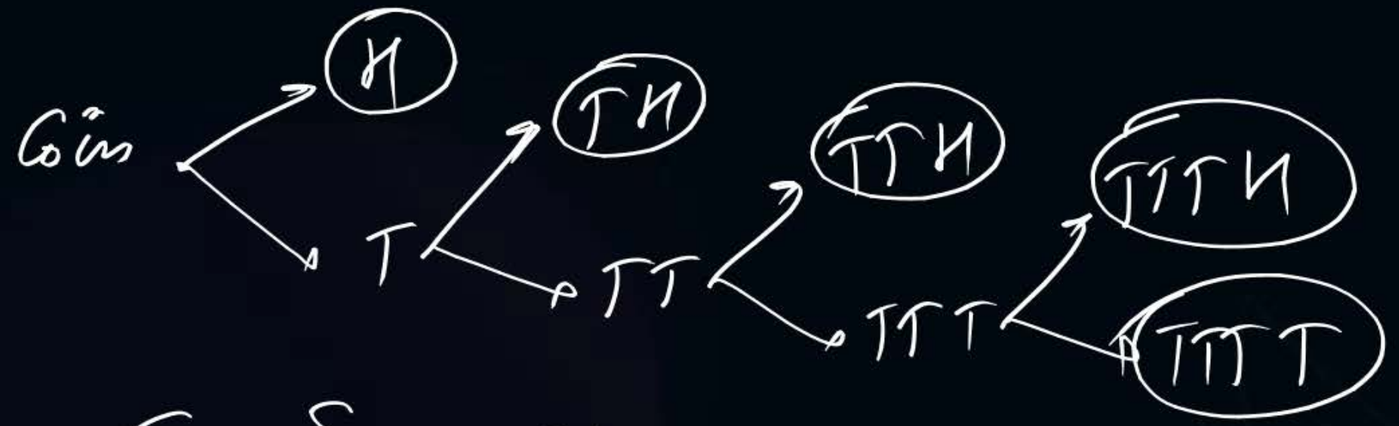
Req Prob $= P[(CS\ CS\ CS) \text{ or } (CS\ M\ CS)]$

$= P[CS\ CS\ CS] + P[CS\ M\ CS]$

$= (1 \times 0.4 \times 0.4) + (1 \times 0.6 \times 0.4)$

$= 0.4 = 40\%$ chance that she will also study CS on Wed.

Note: A coin is tossed (until) Head appears or tail appears 4 times in succession then write it's S-Space?



$$S = \{ H, TH, TTH, TTTH, TTTT \}$$

F.Q. 6 dice are thrown simultaneously then find the prob that,

① All will show the same face = ? $= \frac{fav}{Total} = \frac{{}^6C_1 \times \frac{6!}{6!}}{6^6} = \frac{6 \times 1}{6^6} = \frac{1}{6^5}$

fav Cases = {(111111) (222222) (333333) (444444) (555555) (666666)}

② All will show the different faces = ? $= \frac{fav}{Total} = \frac{6!}{6^6} = Ans$

fav Cases = {eg (123456) .....}

③ Exactly 3 will show the same face & Rest 3 will show different faces = ?

fav Cases = {eg (512554) .....} $= \frac{fav}{Total} = \frac{{}^6C_1 \times {}^5C_3 \times \frac{6!}{3!}}{6^6}$

Explanation of part (3) —

$(1,2,3,4,5,6)$

$(111234), (222134), (333124), (444123), (555123), (666123)$

$\downarrow {}^5C_3$ or $\downarrow {}^5C_3$ or $\downarrow {}^5C_3$ or $\downarrow {}^5C_3$ or $\downarrow {}^5C_3$ or $\downarrow {}^5C_3$

10 Cases, 10 Cases, 10 Cases, 10 Cases, 10 Cases, 10 Cases

$$\text{fav Cases} = 6 \times {}^5C_3 \times \frac{6!}{3!} = 6 \times 10 \times \frac{6!}{3!}$$

Analysis: (Overall Analysis) —

(All diff) or (two same & 4 diff) or (three same & three diff) or (4 same & 2 diff) or (5 same & 1 diff)

or (two same & two same & two diff) or (two same & three same & one diff) or (two same & 4 same)

or (three same & three same) or (All same) = Total (6^6)

→ WRONG App: ~~(All same) or (All diff) = Total (6^6)~~

THANK - YOU